Ref. [60] — Gradient Descent Provably Escapes Saddle Points in the Training of Shallow ReLU Networks

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1. Problem: what if the network is not smoothly differentiable?

Most strict-saddle avoidance theorems assume a twice-continuously differentiable objective. A ReLU network is piecewise linear in its input and generally creates a parameter-space loss that need not satisfy classical global smoothness hypotheses. The authors ask whether a suitably weakened dynamical-systems result can still explain escape near saddle points in concrete shallow ReLU-like training problems.

2. Mathematical setup and intuition

A shallow scalar-input network can be expressed schematically as

$$f_\theta(x) = c + \sum_{j=1}^{m} a_j \sigma(w_j x + b_j), \qquad \sigma(z) = \max\{0, z\},$$

with a square integral loss against an affine target $g(x)$:

$$\mathcal{L}(\theta) = \int \big(f_\theta(x) - g(x)\big)^2 \, \mathrm{d}\mu(x).$$

The exact paper covers specified ReLU and leaky-ReLU parameterizations, target classes and measures; these equations illustrate the problem family rather than quote a single universal model. Since σ has a kink at zero, the update $G(\theta) = \theta - \eta \nabla \mathcal{L}(\theta)$ may fail to satisfy regularity conditions of a standard C^2 stable-manifold theorem.

3. Paper method: weaken the stable-manifold regularity

The authors establish a version of the center-stable manifold theorem with relaxed differentiability requirements and apply it after a detailed structural classification of critical points in shallow scalar-input ReLU and leaky-ReLU training with affine targets. This allows them to prove that GD *circumvents most saddle points* in the analyzed family. They additionally establish global-minimum convergence under favorable initialization criteria expressed using an explicit threshold involving the limiting loss.

> **Principal result.** For the specified shallow ReLU-like problems, saddle avoidance can be proved without assuming that the full training objective obeys the classical smoothness regime; global convergence needs additional conditions on the initialization/limiting loss.

4. An illustrative kink and a key subtlety

For the scalar ReLU $\sigma(z) = \max(0, z)$, the derivative is 0 for $z < 0$ and 1 for $z > 0$, but it is not classically defined at $z = 0$. However, integrating a composed loss over an input distribution can sometimes smooth parts of the objective: *a ReLU activation does not by itself prove that every parameter-space expected loss is everywhere nonsmooth.* That is why the paper analyzes the actual loss and its critical points rather than simply declaring all smooth arguments invalid.

5. What the theorem does not establish

The target functions are affine; the networks are shallow with scalar input; the loss is the square integral loss under specified conditions. Thus the result is not an unconditional statement for arbitrary deep ReLU, CNNs, Transformers, high-dimensional inputs or finite-data cross-entropy. "Most saddles" and "global minima under favorable initialization" must not be collapsed into universal global convergence. No general polynomial saddle-escape complexity is provided.

> **Scope.** This paper closes part of the *smoothness-assumption* gap for a structured shallow ReLU class, not the entire practical-neural-network theory gap.

This makes it valuable to **RQ1**, **RQ2** and especially **RQ4**.

6. Our notes

This is a good counterpoint to the classical strict-saddle papers: it tackles an actual ReLU-like model instead of assuming a globally smooth objective. The authors do not claim every deep ReLU network escapes all saddles. Their theoretical advance is a weaker stable-manifold tool plus a detailed analysis of a very specific shallow training problem, with an extra condition needed for global convergence. **Original source:** https://doi.org/10.1007/s10957-024-02513-3. **Verification:** open-access journal version and paper text; schematic model labelled as illustration.